

- 1 The Sumatran orangutan, *Pongo abelii*, is found only on the island of Sumatra, Indonesia.

Fig. 1.1 shows a Sumatran orangutan.



Fig. 1.1

- (a) The International Union for Conservation of Nature (IUCN) is the world's largest global environmental organisation. The IUCN Red List of Threatened Species™ evaluates the conservation status of plant and animal species. The Sumatran orangutan is categorised as critically endangered on the IUCN Red List.

The Sumatran orangutan spends most of its time in trees and is very sensitive to habitat destruction.

Table 1.1 shows the area of natural forest of Sumatra and the numbers of orangutans in 1985 and 2016.

Table 1.1

	1985	2016
area of natural forest /million hectares	25.3	12.4
number of orangutans	25 000	7300

- (i) With reference to Table 1.1, state the relationship between area of natural forest and number of orangutans.

.....
.....[1]

- (ii) Calculate the mean annual decrease in orangutan numbers between 1985 and 2016.

Show your working.

Give your answer to the nearest whole number.

mean annual decrease =[2]

- (iii) Use your answer from (a)(ii) to estimate the number of years after 2016 that it will take for the Sumatran orangutan to become extinct in the wild, if conservation work is unsuccessful.

..... years [1]

- (b) Suggest **two** reasons for the decrease in numbers of Sumatran orangutans, other than habitat loss.

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.....[2]

- (c) Outline the role zoos can take in the protection of the Sumatran orangutan.

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.....[3]